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Human Habitable Environment Numbers

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ECLSS Targets



- Environmental Control and Life Support Systems (ECLSS)
- These are the numbers your ECLSS system must meet
- Always start designing to the center of any human tolerance bands
- This does not provide the rationale or physiology behind those numbers.





Atmosphere: composition & pressure



- $\geq 30\%$ diluent gas – usually nitrogen
- $\text{ppCO}_2 \leq 3 \text{ mmHg}$ (1-hr avg)
- total pressure 5.0–15.0 psia
- pressure change rate $\leq 13.5 \text{ psi/min}$





Temperature & humidity

- Health limits: 18°C/75%RH to 27°C/25%RH
- Performance zone: ~20–25°C, 30–60%RH; adjustable $\geq 1^\circ\text{C/hr}$

RH – Relative Humidity





Ventilation

- Air velocity 15–120 ft/min to prevent CO₂ or thermal pockets
- must maintain control during off-nominal and maintenance access





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Contamination Control



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- Total dust $<3 \text{ mg/m}^3$, respirable $<1 \text{ mg/m}^3$;
- gaseous pollutants below SMAC limits;
- HEPA-level microbial control;
- Martian dust specifically: 0.1 mg/m^3 , 24-hr TWA — new and stricter than the general dust limit, This is the level to design around!

SMAC – Spacecraft Maximum Allowable Concentration, HEPA – High Efficiency Particulate Air Filter, TWA – Time Weighted Average



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Monitoring & alerting



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- Continuous recording + real-time display of pressure, humidity, temperature, ppO₂, and ppCO₂
- local AND remote alerting when out of bounds
- Note: this is a system requirement, not just a sensor specification



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Water

- 2.5 L per crew per day minimum for hydration
- Potable water chemically + microbiologically safe at point of use
- Hot water band 68–79 °C





Acoustics

- Continuous noise $\leq$ NC-50 (work) / NC-40 (sleep, >30-day missions);
- hazardous ceiling <85 dBA off-nominal

NC-50 / NC-40 — Noise Criterion curves (standardized octave-band sound pressure level limits; higher number = louder allowance)

dBA — A-weighted decibels (sound level scale adjusted for human ear sensitivity)





Key Sources



Summarized information - Please see the handout ECLSS Requirements Mars Habitat 2026

Sources:

- NASA Spaceflight Human-System Standard 3001 Volume 2 Rev E Human Factors, Habitability, and Environmental Health (2025) - <https://www.nasa.gov/ochmo/human-spaceflight-and-aviation-standards/>
- Childress et al. NPJ Microgravity (2023) - <https://www.nature.com/articles/s41526-023-00275-2.pdf>
- Ebert et al. Martian Dust Exposure Permissible Exposure Limits (2026) - <https://www.nasa.gov/general/establishing-crew-exposure-limits-of-martian-dust/>

